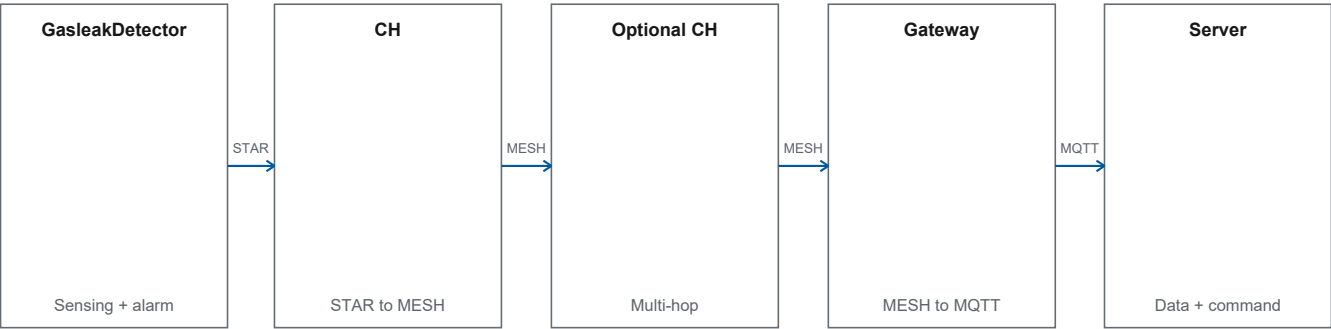


Whole System

Technical Datasheet Revision 4.0

End-to-end GasleakDetector - CH - optional CH - Gateway - broker - Server architecture.



TOPOLOGY	CARRIER	UPLINK	DOWNLINK
Dynamic multi-hop	920-923 MHz	Normal pull + alarm push	Command via broker

Key features

- Primary operational path: GasleakDetector to serving CH, optional transit CH, Gateway, broker, then Server.
- Normal telemetry and alarms use different delivery behavior.
- Commands return through the broker, Gateway, CH route, and endpoint receive window.
- The engineering dataset path is separate from the operational alarm path.

Technical overview

The Whole System connects sensing endpoints, the field radio network, an IP bridge, a broker, and the Server application.

Functional profile

Endpoint	GasleakDetector
Field network	CH STAR/MESH
IP bridge	Gateway
Application	Server

Operational interfaces

GasleakDetector - CH	LoRa STAR Uplink + receive window Telemetry, alarm, and limited downlink
CH - CH/Gateway	LoRa MESH Bidirectional Multi-hop routing and relay
Gateway - network	Wi-Fi STA Bidirectional IP connectivity
Gateway - broker	MQTT / TLS option Bidirectional Publish uplink; subscribe command

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1 Product identification

Product	Whole System
Product status	Engineering Prototype System
Document revision	4.0
Technical baseline	GasleakDetector 0.8.19 CH 0.8.0 Gateway 0.2.0 Protocol 0.2.0
Issuer	Lab IoT ITB

1.1 Abbreviations

Abbreviation	Meaning	Abbreviation	Meaning
AES-GCM	Authenticated encryption on field payloads	MQTT	Broker-based messaging transport
CH	Serving/transit communication node	STAR	GasleakDetector-to-serving-CH radio link
E2E	End-to-end	TLS	Transport security on two separate broker sessions
MESH	CH and Gateway radio backbone	VM	Customer virtual machine for Server

2 System overview

2.1 System scope

The Whole System connects sensing endpoints, the field radio network, an IP bridge, a broker, and the Server application.

Endpoint	GasleakDetector	Field network	CH STAR/MESH
IP bridge	Gateway	Application	Server

- Primary operational path: GasleakDetector to serving CH, optional transit CH, Gateway, broker, then Server.
- Normal telemetry and alarms use different delivery behavior.
- Commands return through the broker, Gateway, CH route, and endpoint receive window.
- The engineering dataset path is separate from the operational alarm path.

2.2 Operational and dataset architecture

The operational path and dataset path have different purposes and processing rules.

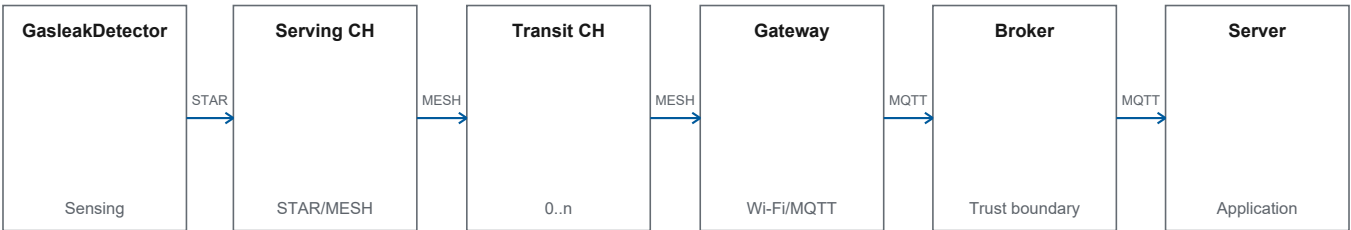


Figure 1. Primary operational path.

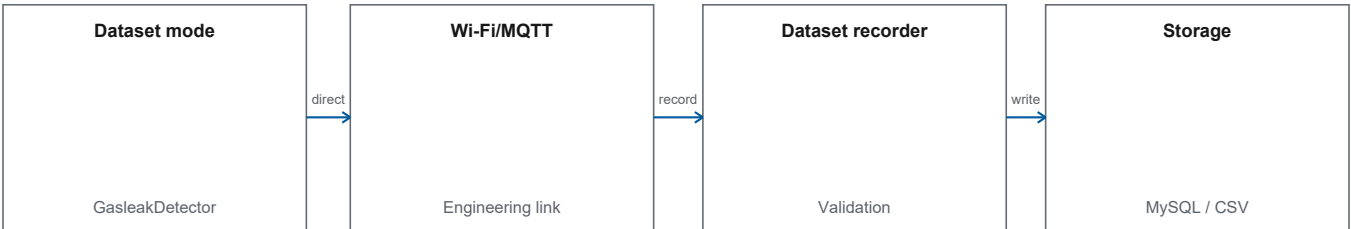


Figure 2. Separate engineering dataset path.

Note: Dataset mode is not the production alarm path and does not replace the operational LoRa path.

2.3 Subsystem responsibility matrix

Each subsystem has a distinct functional boundary across sensing, transport, routing, security, and application.

Subsystem	Primary role	Not responsible for
GasleakDetector	8-channel sensing, local inference, alarm output, LoRa STAR	MESH routing or broker TLS
Serving CH	STAR ingress, cache, MESH forwarding	Payload decryption or Server application
Transit CH	MESH relay and route continuity	Endpoint sensing
Gateway	MESH-to-Wi-Fi/MQTT bridge	Persistent storage or guaranteed delivery
Broker	MQTT session/transport boundary	Field sensing/inference
Server	Validation, decode, alarm routing, topology, command	Physical alarm output

Note: The broker is a logical deployment dependency and must be included in the network design even though it is not a field device.

2.4 Inter-subsystem interface summary

The following table summarizes transport, communication direction, and public function at each subsystem boundary.

Link	Transport	Direction	Function
GasleakDetector - CH	LoRa STAR	Uplink + receive window	Telemetry, alarm, and limited downlink
CH - CH/Gateway	LoRa MESH	Bidirectional	Multi-hop routing and relay
Gateway - network	Wi-Fi STA	Bidirectional	IP connectivity
Gateway - broker	MQTT / TLS option	Bidirectional	Publish uplink; subscribe command
Server - broker	MQTT / TLS	Bidirectional	Ingestion and command
GasleakDetector - local	RS-485 Modbus	Read-only	Local data/status integration

Note: An external Wi-Fi-to-Ethernet converter is not a native Gateway interface and is not included in the product matrix.

3 Operational flows

3.1 Normal telemetry sequence

Normal telemetry uses a cache at the serving CH and is sent in response to a server pull.

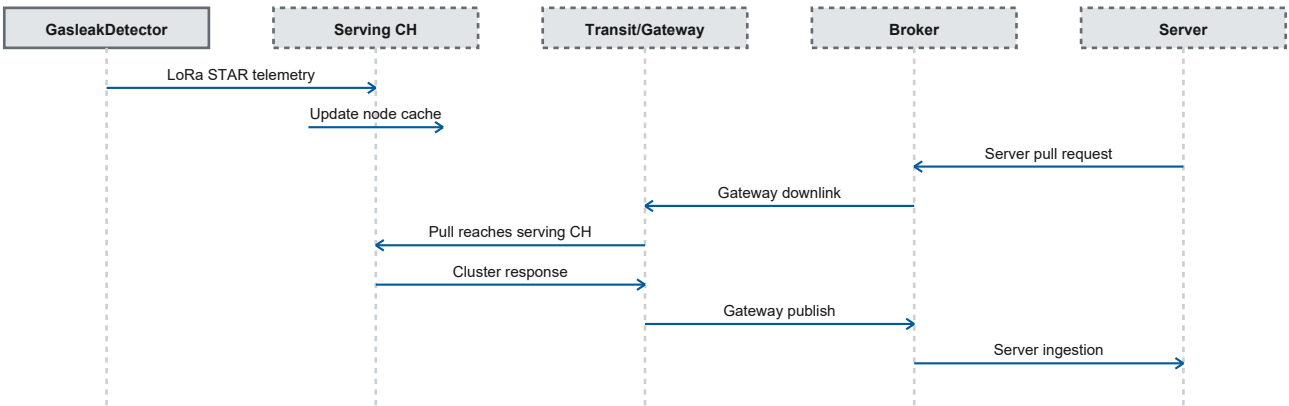


Figure 3. Cache/pull normal telemetry.

Note: Normal telemetry delivery is pull-based and follows the data available in the serving-CH cache.

3.2 Alarm and return-to-clear sequence

Alarm/clear uses a push path and becomes a Server alarm only after the payload passes the integrity gate.

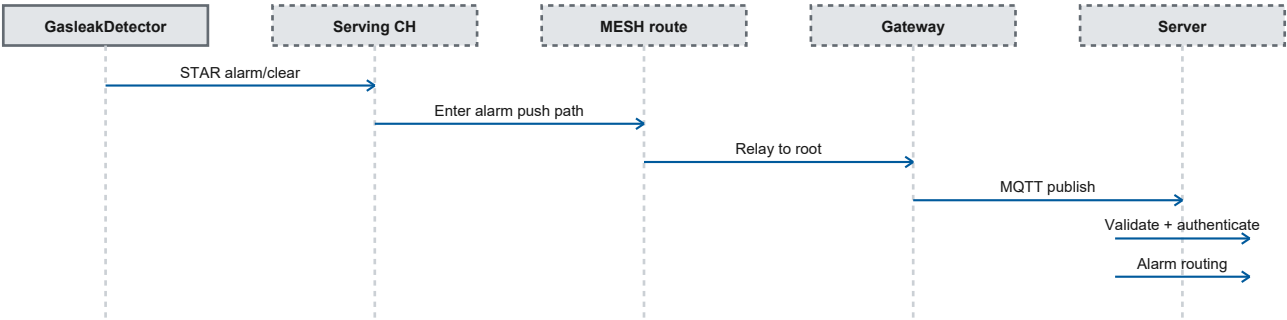


Figure 4. Alarm push and Server integrity gate.

Local alarm	Steady 24 V output in GasleakDetector AUTO mode.
Network alarm	Authenticated event routed by the Server.

3.3 Command downlink sequence

A high-level mode command moves from the Server through the broker and field route to the GasleakDetector receive window.

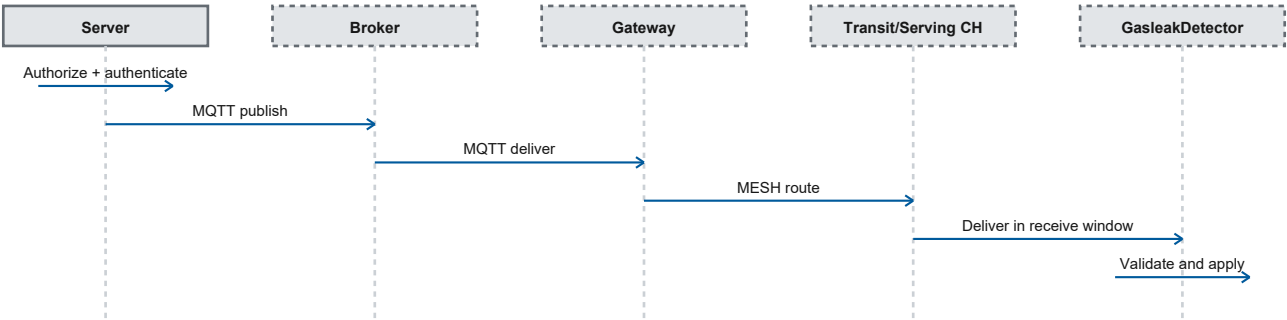


Figure 5. Command downlink to the endpoint.

Note: Command delivery to battery endpoints occurs during the endpoint post-transmission receive window.

4 Routing and RF

4.1 Discovery and route formation

CH forms routes dynamically from candidates with lineage toward the Gateway root.

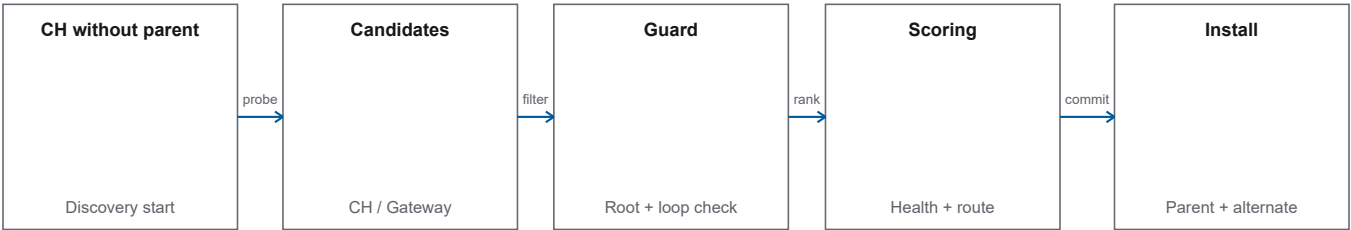


Figure 6. Discovery through parent installation.

- Gateway is the root destination, not every CH's static parent.
- The lineage guard prevents routing loops.
- The actual parent is not fixed to a specific identity.

4.2 Multi-hop and failover

An alternate parent and rediscovery maintain continuity when the active parent is no longer healthy.

Stage	Description
Healthy route	Serving CH -> transit(s) -> Gateway
Health failure	Timeout/HELLO ACK/alarm ACK failure
Alternate parent	Used when valid
Rediscovery	Builds new candidates
Recovered route	New parent follows discovery

Figure 7. CH route failover.

Note: Multi-hop and failover have been validated at the ITB IoT Lab. The replacement parent follows discovery and is not fixed to a specific identity.

4.3 Two-domain RF plan

STAR and MESH share the 920-923 MHz accepted carrier range but use different defaults and antennas.

Domain	Default	BW / SF / CR	TX setting	Antenna
STAR	920.0 MHz	125 kHz / SF7 / 4/5	17 dBm	3 dBi
MESH	921.0 MHz	125 kHz / SF9 / 4/5	17 dBm	8 dBi
Accepted carrier		920-923 MHz inclusive		
Active configuration		Verified through device readback.		

Note: Nominal calculation before loss: STAR 20 dBm and MESH 25 dBm.

5 Networking and security

5.1 IP network, broker, and TLS profile

When the TLS Gateway profile is selected, the Gateway and Server each establish their own TLS session to the broker; the broker is the termination/trust boundary.

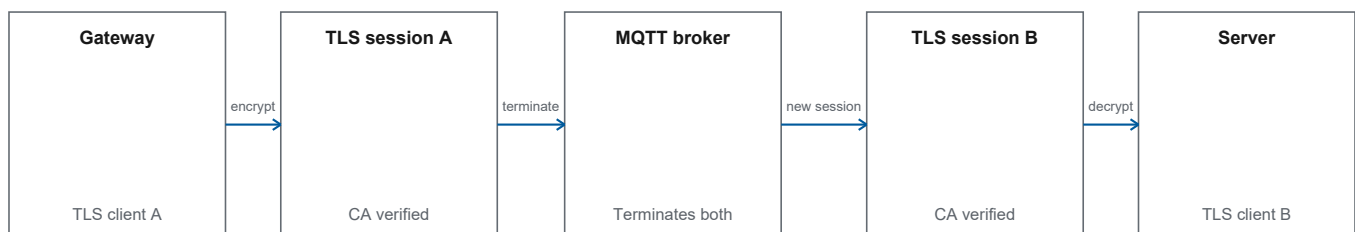


Figure 8. Two TLS sessions in the TLS deployment profile.

- The TLS Gateway profile requires a Root CA and a system epoch that meets the trusted-time threshold.
- The Server flow supports credentials and CA-verified TLS to the broker.
- In the TLS profile, the Gateway and Server use two separate TLS sessions, each terminating at the broker.

5.2 Layered security boundaries

Payload security and transport security protect different layers and segments.

Layer	Coverage	Function	Termination
AES-128-GCM	GasleakDetector payload	Confidentiality + authentication	Server decoder
Gateway TLS	Gateway - broker	MQTT transport protection	Broker
Server TLS	Server - broker	MQTT transport protection	Broker
Command auth	High-level downlink	Authorization + payload authentication	Target application

Note: CH and Gateway forward the encrypted field payload; it is decrypted/authenticated at the Server.

5.3 Availability, cache, and buffering

Temporary data resides in the CH cache or Gateway RAM queue; field retention follows each mechanism's capacity and memory type.

Location	Mechanism	Persistence	Boundary
CH	Node cache for normal telemetry	Runtime memory/state	Sent on server pull
Gateway	8-item MQTT queue, 1-1023 B payload	Volatile RAM	New item rejected when full
Server	Anti-replay state	Persistent storage required in deployment	Reloaded after service restart
Dataset	MySQL / CSV	Engineering persistence	Dataset records only

6 Power, variants, and deployment

6.1 Subsystem power domains

Each device has a different power boundary; one PCB's capability is not generalized to another subsystem.

Subsystem	Configuration	Monitoring behavior
GasleakDetector	24 VDC nominal; 7 x parallel 18650 for battery mode	Diagnostic voltage monitoring
CH	1-3 x parallel 18650; 2 x parallel 6 W panels	VBAT read-only

Note: CH firmware reports VBAT read-only without active low-voltage cutoff. GasleakDetector also monitors voltage without active undervoltage cutoff.

6.2 Hardware variants and firmware compatibility

CH and Gateway have Rectangle/Circle boards; Gateway also has TLS/non-TLS choices.

Device	Board	Transport option	Firmware target
GasleakDetector	Product board	LoRa STAR + RS-485	GasleakDetector firmware
CH	Rectangle	STAR + MESH	CH Rectangle
CH	Circle	STAR + MESH	CH Circle
Gateway	Rectangle	MQTT non-TLS / TLS	Gateway Rectangle - standard / TLS
Gateway	Circle	MQTT non-TLS / TLS	Gateway Circle - standard / TLS
Server	Customer VM	MQTT client/application	Deployment-specific

Note: After upload, device readback verifies the active board and transport variant.

6.3 Deployment topology

Deployment requires field devices, Wi-Fi, a broker, a Server VM, optional dataset database, and operational ownership.

Stage	Description
Field layer	GasleakDetector, serving/transit CH, Gateway
Site network	Wi-Fi, DNS, NTP, VLAN/NAC, firewall
Messaging	MQTT broker, CA chain, ACL, credentials
Application	Node-RED Server on customer VM
Engineering data	Dataset recorder and MySQL/CSV when used

Figure 9. System deployment layers.

Note: OS image, VM sizing, broker, database, backup, HA, RBAC, and SSO are specified for each customer deployment.

6.4 Commissioning sequence

Commissioning must sequentially verify firmware, identity, radio, network, security, and the application path.

Stage	Description
1. Firmware	Select exact device/board/transport target
2. Device readback	Version, identity, board variant, RF configuration
3. Field link	STAR, MESH, discovery, parent, route
4. IP/TLS	Wi-Fi, DNS/NTP, CA, broker connection
5. Application	Decode, alarm route, topology, command
6. Record	Logs, unit IDs, versions, and pass criteria

Figure 10. Engineering commissioning sequence.

Note: Each commissioning stage is completed in sequence and recorded in the deployment acceptance result.

7 Integration contract and system limits

7.1 System interface contract

Each subsystem boundary uses a fixed communication profile so that the GasleakDetector, CH, Gateway, and Server can be integrated consistently.

Boundary	Interface	Profile
GasleakDetector - CH	LoRa STAR	920.0 MHz; SF7; BW 125 kHz; CR 4/5; TX power setting 17 dBm; antenna 3 dBi
CH - CH / Gateway	LoRa MESH	921.0 MHz; SF9; BW 125 kHz; CR 4/5; TX power setting 17 dBm; antenna 8 dBi
Gateway - MQTT broker	Wi-Fi STA / MQTT	MQTT over TLS on the TLS profile; certificate verification and time synchronization are mandatory.
Server - MQTT broker	MQTT	A separate Server-to-broker TLS session uses customer-deployment authentication.
Server - Database	MySQL	Engineering-dataset storage with idempotency checks.
GasleakDetector - controller	RS-485 / Modbus RTU	Unit 1; 9600 bit/s; 8N1; FC03/FC04; 8 read-only registers

Note: The product operating band is limited to 920-923 MHz. The nominal STAR and MESH frequencies are within this band.

7.2 Subsystem capacity limits

The following capacities are implementation limits to consider when designing node counts, downlink flows, and connection recovery.

Subsystem	Resource	Limit
GasleakDetector	Sensor channels	8
GasleakDetector	Modbus registers	8 read-only registers
CH	Parent candidates	8
CH	Node cache	32 entries
CH	Alarm / transmission queue	8 / 8 entries
CH	Pending downlink	16 entries
Gateway	MQTT publication queue	8 volatile messages
Gateway	Deferred-publication payload	1023 bytes maximum
Server	Sensor values per dataset record	8

7.3 Failure behavior

Each subsystem response to a disturbance is explicit so that recovery and data-retention limitations can be considered in deployment design.

Disturbance	System response	Limitation
CH parent no longer eligible	The CH performs discovery and selects a parent candidate that satisfies the current rules.	The replacement parent is not statically assigned.
Wi-Fi or broker unavailable	The Gateway retries publication and uses the local queue when an item can be admitted.	The queue is volatile and holds at most eight items.
TLS prerequisite incomplete	The Gateway blocks the MQTT connection.	There is no fallback to a TLS transport without verification.
MySQL storage unavailable	The dataset recorder continues writing the CSV copy.	MySQL status must be monitored and reconciled by the server operator.
Invalid dataset record	The Server rejects the record before storage.	The record must satisfy every admission rule.
Low battery voltage	The GasleakDetector and CH report the monitoring result.	Firmware does not automatically disconnect power or enter low-power mode based on this threshold.

7.4 Security boundaries

End-to-end security is formed by radio controls, application-message validation, Gateway-to-Server transport, and Server infrastructure controls.

Boundary	Control	Operational responsibility
Field radio hop	Node identity, message sequence, length, and CRC are checked by the radio application protocol.	Device identities and radio configuration are managed during provisioning.
Server application	The Server performs AES-GCM authentication and replay rejection; persistent state is required in deployment.	Security material and anti-replay storage are managed in the Server deployment.
Gateway / Server - broker	When the TLS Gateway profile is selected, the Gateway and Server use two separate MQTT-over-TLS sessions with certificate validation.	The customer supplies the CA, time endpoint, broker, and authentication policy.
Server services	Credentials reside in the runtime environment or a secret store; persistent anti-replay storage must be configured.	The customer manages network access, service accounts, backup, patching, and VM monitoring.
Storage	A unique record identifier prevents dataset duplication in primary storage.	Retention, backup, and reconciliation are defined in customer operating procedures.

8 References and revision history

Component and protocol documents are used as system-specification references.

Reference	Document
R1	GasleakDetector Technical Datasheet, Rev 4.0
R2	CH Technical Datasheet, Rev 4.0
R3	Gateway Technical Datasheet, Rev 4.0
R4	Server Technical Datasheet, Rev 4.0
R5	Firmware/Protocol baseline: 0.8.19 / 0.8.0 / 0.2.0 / protocol 0.2.0
Rev	Change
3.0	Updated technical structure and specifications.